Ahmed Abo-Shadi

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Education

- **California State University, Fullerton** – MS in Computer Engineering

- **University of California, Irvine** – BS in Software Engineering
  - Completed with a GPA of 3.671.

Listed Skills

- Python
- Rust
- C++
- Luau

- Reverse Engineering
- Binary Formats
- TCP/UDP
- GNU/Linux
- Bash

- Git/GitHub
- React
- Next.js
- Sqlite

Employment

- **Engineering Assistant @ Professional Engineering Center** – June 2015 thru present
  - Managing high-impact private data going back since 2003, enabling reliable file access for the engineering team.
  - Reverse-engineered and modified legacy plugins for compatibility with modernA utoCAD releases, extending an established software stack without replacing the underlying workflow.
  - Translating sketches into production-ready structural designs for commercial and industrial projects in high-stakes industries.

- **Frontend Developer at GameIn, UCI Capstone** – Sep 2023 thru Mar 2024
  - Implemented mobile-user-friendly pages; leveraged Next.js (and thus React) from mockups that we also made.
  - Arranged, as part of a team, a slide deck, using Figma for interface design.

Projects & Research

- **Responsible Security Disclosure - YuJa API** – Dec 2022
  - Responsibly disclosed an API access-control vulnerability involving 6,000,000+ private academic recordings.
  - YuJa released a security patch after ~2 days.

- **Roblox Freedom Distribution** – July 2023 thru present
  - Built a full-stack Python-based bootstrapper and a client-server system for self-hosted Roblox game instances, including TCP and UDP networking, an HTTPS web server, configurable ports and automated remote-asset serialization.
  - Designed the system to support multiple software versions; the project has 150+ stars on GitHub.
  - Published detailed code-analysis guides demonstrating quirks of the Roblox software.

- **RBXL Version Forensics** – May 2025 thru May 2026
  - Built a Rust command-line tool that parses binary `rbxl` files and infers the Roblox Studio version from structural properties.
  - Benchmarked the implementation against 10,285 binary files in approximately 4.5 minutes, using a modified binary deserializer to inspect property names without parsing full property values.